llmXive Automated Review of Gemma 4 Technical Report

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a generally coherent technical report for the Gemma 4 family, but several specific claims regarding performance metrics and architectural ratios require clarification to ensure they are fully supported by the provided tables and text.

First, there is a minor inconsistency in the description of the attention mechanism ratios. The Introduction states a "5:1 ratio of local sliding window to global self-attention" for the model suite, while Section 2.1 clarifies that this 5:1 ratio applies to models *except* E2B, which uses a 4:1 ratio. The introductory claim should be qualified to exclude E2B to align with the detailed architecture section.

Second, the parameter reporting in Table 1 versus the text in Section 2 creates potential ambiguity. The text explicitly distinguishes between "effective" parameters (2.3B/4.5B) and "total" parameters (5B/8B) for E2B/E4B due to per-layer embeddings. However, Table 1 lists specific numbers (e.g., 2,340M embedder parameters) without explicitly labeling the column as "effective" or "total" in a way that immediately resolves the 2.3B vs 5B distinction for a quick reader. Clarifying the table headers or caption would prevent misinterpretation of the model sizes.

Third, the performance comparison in Section 5.1 regarding Codeforces Elo appears contradictory to the data in Table 2. The text claims E2B "roughly matches" Gemma 3 27B, yet the table shows E2B at 633 Elo and Gemma 3 27B at 110 Elo. A 6x improvement is not a "match." This is likely a phrasing error where the authors meant to highlight the efficiency gain (10x fewer parameters) rather than a performance match, or they intended to compare against a different baseline. The claim needs correction to accurately reflect the significant performance lead shown in the table.

Finally, while the claim that Gemma 4 "rivals" larger models is supported by the Elo leaderboard (Table 3), the specific phrasing in the Introduction and Section 5.1 could be slightly refined. The 31B model is the top *dense* open model, but it trails the top *MoE* open models (e.g., GLM 5.1) by a noticeable margin (1451 vs 1475). Ensuring the text specifies "rivals larger dense models" or "competes with frontier models" without implying parity with the absolute top MoE scores would be more precise.

These issues are primarily matters of precise wording and table clarity rather than fundamental scientific flaws, but they are necessary for the reader to trust the specific comparative claims made.

Action items.

- [**writing**] The paper presents a generally coherent technical report for the Gemma 4 family, but several specific claims regarding performance metrics and architectural ratios require clarification to ensure they are fully supported by the provided tables and text. First, there is a minor inconsistency in the description of the attention mechanism ratios. The Introduction states a "5:1 ratio of local sliding window to global self-attention" for the model suite, while Section 2.1 clarifies that this 5:1 rati

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively visualizes the MTP drafter architecture and its connection to the main model, but the caption's mention of 'KV cache' is not explicitly labeled in the diagram, and the input variable 'p1' is undefined.

Figure 2

The figure illustrates image resizing but contains a significant contradiction between the visual grid (3x3) and the caption's description (2x4). Additionally, the caption includes a formatting error ('pooled 33') that obscures the technical explanation.

Action items.

- [**writing**] Figure 1: The caption states the drafter is fed 'activations and KV cache', but the diagram only explicitly labels 'Last Prefilled Global Layer' and 'Last Prefilled Local Layer' as inputs; the specific 'KV cache' input is not visually labeled.
- [**writing**] Figure 1: The input labels at the bottom ('t1 / p1', 't2 / p1', 't3 / p1') are undefined in the caption, making it unclear what 'p1' represents in the context of the autoregressive process.
- [**science**] Figure 2: The caption claims the image is resized to '2 x 4 pooled patches', but the visual grid clearly displays 3 rows and 3 columns (9 patches). This contradicts the text description of the pooling layout.
- [**science**] Figure 2: The caption states the image is resized to '2 x 4 pooled patches (each of size 48px^2)', but the visual grid shows 9 patches, and the math for a 96x192 image with pooling kernel 3 and patch size 16 does not yield 48px^2 pooled patches in a 2x4 arrangement.
- [**writing**] Figure 2: The caption contains a typo 'pooled 33' which appears to be a formatting error or missing text, making the sentence 'the vision encoder representations are pooled 33' illegible.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured for a technical audience, but it relies on several acronyms and specific notation conventions that are introduced without explicit definition, creating minor barriers for a competent reader from an adjacent field (e.g., a computer vision or speech processing researcher not deeply embedded in the specific LLM subfield).

Specifically, the acronyms "MTP" (Multi-Token Prediction), "QAT" (Quantization-Aware Training), "PT", and "IT" are used in the Introduction and early sections before being spelled out. While these are standard in the immediate LLM community, the "jargon police" lens requires them to be defined at first use for the broader adjacent-field PhD. Similarly, "p-RoPE" is a specific variant of Rotary Positional Embeddings; while the citation is provided, a brief parenthetical explanation of what the "p" denotes (e.g., "power" or "position") or a one-sentence gloss would prevent the reader from having to cross-reference the citation to understand the core mechanism.

The model naming convention "E2B" and "E4B" is also used heavily without a clear prose definition of the "E" prefix (Effective) in the main text, relying on the table caption for clarification. This forces the reader to flip between the text and the table to understand the parameter counts. Finally, the dimensionality of the audio vectors (640) is stated as a fact without explicitly linking it to the 40ms chunk size and 16kHz sampling rate in the immediate sentence, which is a small but necessary operational detail for reproducibility and understanding.

These are all low-effort fixes (adding a parenthetical or a short clause) that would significantly improve the self-containment of the paper for the target "adjacent-field" audience.

Action items.

- [**writing**] Section 1 (Introduction) uses 'p-RoPE' and 'KV cache' without definition. While 'KV cache' is common in LLM literature, 'p-RoPE' is a specific variant introduced in a cited paper; define it at first use (e.g., 'p-RoPE (a variant of Rotary Positional Embeddings)') to ensure adjacent-field readers understand the specific modification.
- [**writing**] Section 1 (Introduction) and Section 2 use 'MTP' (Multi-Token Prediction) and 'QAT' (Quantization-Aware Training) as acronyms before they are explicitly defined in the text. Expand these at their first occurrence in the Introduction or early in Section 2 (e.g., 'autoregressive multi-token prediction (MTP) drafter head').
- [**writing**] Table 1 caption and Section 2 use 'E2B' and 'E4B' as model identifiers without defining the 'E' prefix or the numbers in the main text. While the table caption mentions 'effective', the specific naming convention (Effective 2B/4B) should be explicitly defined in the prose when first introduced to avoid confusion with total parameter counts.
- [**writing**] Section 2.1 (Vision modality) introduces '2D-RoPE' and '2D absolute positional embeddings' without a brief gloss. While 'RoPE' is standard, the '2D' adaptation for vision is a specific architectural choice; a short clause explaining that this extends rotary embeddings to 2D spatial coordinates would aid adjacent-field readers.
- [**writing**] Section 2.3 (Encoder-free architecture) mentions '48x48x3 RGB patches' and '640-

dimensional vectors' without defining the source of the 640 dimension (e.g., '640-dimensional vectors (derived from 16kHz audio sampled at 40ms chunks)'). The dimensionality is critical for understanding the projection but is currently implicit.

- [**writing**] Section 2.4 (Pre-training) and Section 3 (Instruction Tuning) use 'PT' and 'IT' as acronyms for 'Pre-training' and 'Instruction Tuning' without defining them. These are common in the field but should be expanded at first use (e.g., 'Pre-training (PT) versus Instruction Tuning (IT) formatting') for clarity.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a generally coherent argument for the Gemma 4 model family, with a clear progression from architecture to evaluation. However, there are minor inconsistencies in terminology and numerical reporting that require clarification to ensure the logical flow is unambiguous.

First, the description of the "encoder-free" architecture in the Abstract and Introduction (Section 1) states the model "ingests raw audio and image patches." In contrast, Section 2.3 ("Encoder-free architecture") clarifies that the model uses "lightweight projection modules" (a 35M parameter matmul for vision) and projects "640-dimensional vectors" for audio. While "encoder-free" correctly implies the absence of heavy pre-trained encoders (like the 550M ViT), the phrasing "ingests raw... patches" in the high-level summary suggests a direct input to the LLM that bypasses the projection step described in the body. This is a minor semantic gap; the body is technically correct, but the abstract/intro should be refined to state that raw inputs are processed via lightweight projection modules rather than "ingested" directly, to avoid implying a lack of transformation.

Second, there is a numerical discrepancy regarding the MoE model's active parameters. Table 1 lists the "26B-A4B*" model with "2,800M (active)" Einsums. However, Section 2 ("Model Architecture") explicitly states the MoE model has "3.8B activated parameters." The difference between 2.8B and 3.8B is substantial (approx. 1B parameters). Unless "Einsums" in Table 1 refers strictly to a subset of weights (e.g., excluding biases, embeddings, or specific layer types) while "activated parameters" includes all, this is an unexplained inconsistency. The authors should either align these numbers or clarify the definition of "Einsums" in the table caption to explain why it differs from the stated activated parameter count.

Finally, the definition of $N_{\max}$ in Section 2.1 is ambiguous. The text states, "We restrict the maximum number of tokens, $N_{\max}$ to the values 70, 140, 280, 560 and 1120." Given the context of the paragraph (Vision modality), these values likely refer to the maximum number of *vision* tokens. However, the paper later reports long-context benchmarks (Table 3) with context lengths of 32k and 128k. Without an explicit distinction, a reader might infer that $N_{\max}$ limits the *total* sequence length to ~1,120 tokens, which would contradict the long-context results. The text should explicitly state that $N_{\max}$ refers to the maximum number of *vision* tokens, distinct from the total sequence length supported by the model.

These issues are primarily matters of precision and clarity rather than fundamental logical breaks, but resolving them is necessary for the paper's internal consistency.

Action items.

- [**writing**] The paper presents a generally coherent argument for the Gemma 4 model family, with a clear progression from architecture to evaluation. However, there are minor inconsistencies in terminology and numerical reporting that require clarification to ensure the logical flow is unambiguous. First, the description of the "encoder-free" architecture in the Abstract and Introduction (Section 1) states the model "ingests raw audio and image patches." In contrast, Section 2.3 ("Encoder-free architecture")

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper generally aligns its claims with evidence, but the Abstract and Introduction contain rhetorical overreach regarding the scope of performance gains and architectural novelty.

First, the Abstract asserts the suite "establishes a leap in performance across STEM... benchmarks." However, Table 1 (it_fs) reveals the 12B model scores only 5.2 on HLE (Humanity's Last Exam), a rigorous STEM benchmark, compared to 19.5 for the 31B. This indicates the "leap" is not uniform across the parameter range, particularly on the most difficult reasoning tasks. The claim should be hedged to reflect that significant gains are observed primarily in the larger models or across "most" benchmarks, rather than universally.

Second, the Abstract claims the models "rivals larger, frontier open models in human-rated tasks." Table 2 (lmsys_elo_leaderboard) places the Gemma 4 31B at rank 43 with an Elo of 1451. While competitive, it trails the top open models in the table (e.g., GLM 5.1 at 1475). The phrasing suggests parity with the absolute state-of-the-art, which the data does not fully support. A more precise phrasing would be "rivals specific larger open models" or "achieves competitive performance with a subset of larger models."

Finally, the Introduction highlights the "unified, encoder-free architecture" as a key feature of the "Gemma 4 model suite." However, Table 1 and Section 2.3 confirm this architecture is exclusive to the 12B model; the 2.3B, 4.5B, and 31B models rely on frozen external encoders. Framing this as a suite-wide characteristic without immediate qualification creates a scope mismatch. Explicitly restricting this claim to the 12B model in the Abstract and Introduction would improve accuracy.

These issues are primarily matters of rhetorical precision. The data supports the corrected, narrower claims, and no new experiments are required.

Action items.

- **[writing]** Abstract claims a 'leap in performance across STEM' benchmarks, but Table 1 shows the 12B model scores only 5.2 on HLE (a hard STEM benchmark) vs 19.5 for the 31B. The 'leap' is not uniform across sizes. Narrow the claim to 'improvements in larger models' or 'across most benchmarks'.
- **[writing]** Abstract states the model 'rivals larger, frontier open models.' Table 2 shows Gemma 4 31B (Elo 1451) ranks 43rd, trailing top open models like GLM 5.1 (1475). The claim implies parity with leaders, which data doesn't support. Qualify to 'rivals specific larger models' or cite the specific Elo range.
- **[writing]** Introduction frames 'encoder-free architecture' as a suite feature, but Table 1 shows it applies only to the 12B model; others use frozen encoders. Restrict the claim in the Abstract and Intro to the 12B model to avoid implying a family-wide shift.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a technical report for the Gemma 4 model family, an open-weight, multimodal LLM. From a safety and ethics perspective, the manuscript adequately addresses the primary risks associated with releasing such a system.

Section 7 ("Responsibility, Safety, Security") provides a comprehensive overview of the safety governance, train-time mitigations (data filtering for PII, toxic content, and unsafe utterances), and evaluation protocols. The authors explicitly state that models were evaluated without safety filters to measure inherent capabilities and report that policy violations were minimal. The "Ethical Considerations and Risk Mitigation" subsection (7.4) correctly identifies key areas of concern: bias/fairness, misinformation, and privacy, offering standard guidance for downstream developers (e.g., continuous monitoring, adherence to local regulations).

The paper does not exhibit specific, unmitigated dual-use risks that are unique to this work compared to the broader class of frontier LLMs. The "thinking mode" and "encoder-free architecture" are described as efficiency and reasoning improvements, not as mechanisms designed to bypass safety filters or deceive users covertly. The release of quantized models and drafters is standard for the field and is accompanied by the same safety policies as the base models.

No human-subjects data requiring IRB approval is described; the human evaluations mentioned (Arena) are standard third-party benchmarks where the paper reports aggregate Elo scores rather than raw personal data. The training data is described as a filtered collection of web documents and public sources, with no indication of license violations or the release of PII.

As this is a third-party preprint, the absence of a specific "broader impacts" essay is not a defect, and the existing safety section meets the disclosure standards for a technical report of this nature. There are no fatal gaps, missing disclosures, or specific actionable risks identified that would prevent the paper from being accepted in this lens.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a comprehensive technical report for the Gemma 4 family, but the experimental design in several key tables fails to rule out alternative explanations for the reported performance gains, specifically regarding variance, confounding variables, and baseline fairness.

First, the primary evidence for the "leap in performance" (Abstract, Section 5) relies on single-point numbers in Tables 2 and 3 (e.g., MMLU Pro 85.2, AIME 89.2). There is no reporting of standard deviation, confidence intervals, or the number of random seeds used for training or evaluation. In large language model benchmarks, performance can fluctuate significantly based on initialization and sampling seeds. Without variance metrics, a reader cannot distinguish a genuine architectural improvement from a lucky seed or a specific test-set artifact. The design requires multi-seed reporting to support the claim of a robust "leap."

Second, Table 2 introduces a significant confound in the comparison between Gemma 4 and Gemma 3. The caption explicitly states that the Gemma 3 27B baseline is "non-thinking," while all Gemma 4 results are reported "in thinking mode." The paper attributes the performance gains to the new architecture and training recipe, but the design does not isolate the contribution of the "thinking mode" (reasoning traces). It is plausible that the observed gains are primarily driven by the reasoning capability rather than the underlying model architecture. To support the claim that the *architecture* is superior, the authors must either report Gemma 4 in non-thinking mode or Gemma 3 in thinking mode to isolate the variable of interest.

Third, the claim of an "encoder-free" architecture for the 12B model (Section 2.3) lacks a direct ablation. The paper asserts that replacing the 550M vision encoder with a lightweight projection module maintains performance, but Table 2 does not provide a control run comparing the encoder-free 12B against a 12B model trained with the standard frozen encoders. Without this comparison, the evidence cannot rule out that the performance is due to the encoders being effectively present in the training data or that the "encoder-free" label is a misnomer for a different architectural choice that wasn't fully isolated.

Finally, the long-context benchmarks (Table 4) show significant gains, but the evaluation setup does not explicitly detail whether the baselines were tuned for the same context lengths or if the "thinking mode" was active for all models in these specific tests. If the baselines were evaluated without thinking mode or with different context-window optimizations, the comparison is unfair. The design needs to ensure that the only variable changing between the proposed method and the baseline is the specific architectural component being claimed.

These issues do not invalidate the results but prevent the current evidence from definitively establishing the specific causal claims made about the architecture's superiority over the baseline.

Action items.

- **[writing]** The paper presents a comprehensive technical report for the Gemma 4 family, but the experimental design in several key tables fails to rule out alternative explanations for the reported performance gains, specifically regarding variance, confounding variables, and baseline fairness. First, the primary evidence for the "leap in performance" (Abstract, Section 5) relies on single-point numbers in Tables 2 and 3 (e.g., MMLU Pro 85.2, AIME 89.2). There is no reporting of standard deviation, confiden

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this technical report is largely descriptive, presenting point estimates for model performance without the necessary inferential machinery to support claims of superiority or stability.

Missing Uncertainty Reporting: The most pervasive issue is the reporting of single-point estimates for benchmark results in **Table 2** (Static benchmarks) and **Table 3** (Vision benchmarks). For instance, the 31B model is reported as achieving "85.2" on MMLU Pro and "89.2" on AIME 2026. There is no indication of the number of random seeds used, nor is there any measure of variance (standard deviation, standard error, or confidence intervals). In the context of large language models, where performance can fluctuate based on initialization, sampling temperature, or data shuffling, a single number is statistically insufficient to claim a "leap in performance." The authors must either report mean ± SD over multiple seeds (standard practice) or explicitly state that these are single-run results, which would necessitate softening claims of statistical significance.

Unsubstantiated Claims of Significance: In **Section 5.2**, the text states: "Gemma 4 31B is closest in size and significantly better across the board." The term "significantly" implies a statistical hypothesis test was performed. However, **Table 2** only provides raw scores. No p-values, test statistics, or effect sizes are reported to validate this claim. Without a formal test (e.g., a paired t-test across seeds or a bootstrap test), the word "significantly" is misleading. The authors should either run the appropriate tests and report the results or rephrase the claim to "higher performance" based on the observed point estimates.

Ambiguous Confidence Intervals: Table 1 (Arena Elo leaderboard) reports 95% confidence intervals (e.g., "± 8" for Gemma 4 31B). While this is a positive step, the methodology for deriving these intervals is absent. Elo ratings are typically derived from a Bradley-Terry model or similar pairwise comparison framework. The width of the CI depends heavily on the number of battles and the specific statistical model used. Without specifying the method (e.g., "95% CI derived from 10,000 bootstrap samples of the pairwise comparison matrix"), these intervals are not reproducible or verifiable.

Recommendation: The paper requires a revision to the statistical reporting section. Specifically:

1. Add standard deviations or confidence intervals to all benchmark tables (Tables 2, 3, 4, 5) based on multiple seeds.
2. Remove the word "significantly" from claims unless accompanied by a reported p-value and test name.
3. Add a brief methodological note in the caption of Table 1 explaining how the Elo confidence intervals were calculated.

These are reporting fixes (writing) that can be implemented if the raw per-seed data exists, or require re-running experiments (science) if the data was not collected. Given the scale of the models, it is highly likely the data exists but was omitted for brevity.

Action items.

- [**writing**] Table 2 reports single-point benchmark scores (e.g., '85.2' on MMLU Pro) without any measure of uncertainty (SD, SE, or CI) or the number of seeds/runs used. In ML, single-run reporting is insufficient to distinguish signal from noise. Report mean $\pm$ SD over 3 seeds or explicitly state the result is from a single run and drop the implied precision.
- [**writing**] Table 1 (Arena Elo) reports 95% CIs (e.g., '$\pm$ 8') for Gemma 4 models but provides no methodology for their calculation (e.g., bootstrap, Bayesian inference) or the number of pairwise comparisons used to derive them. Without this, the intervals cannot be verified or interpreted correctly.
- [**writing**] Section 5.2 claims Gemma 4 is 'significantly better' than Gemma 3 across benchmarks based on Table 2 point estimates. No statistical test (e.g., paired t-test, bootstrap) or p-values are reported to support this claim. Replace 'significantly better' with 'higher' or report the actual test statistics and p-values.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper generally presents a clear narrative of the Gemma 4 model family, with a logical flow from architecture to training and evaluation. However, several sections suffer from structural issues that force the reader to re-parse sentences to identify the main point.

In Section 2.1, the description of long-context efficiency is buried in a long, complex sentence that lists multiple architectural choices (sliding window ratios, p-RoPE, KV sharing) before stating the result (37.5% reduction). This "list-then-conclusion" structure creates a garden-path effect. Splitting this into two sentences—one for the methods and one for the result—would significantly improve readability.

Similarly, Section 2.3 lacks a strong topic sentence. The paragraph jumps immediately into specific details about the 12B model's vision processing without first stating the high-level architectural shift (the move to an encoder-free paradigm). A clear opening sentence summarizing the change would help orient the reader before diving into the specific projection details.

In Section 3.1, the human evaluation results are presented with the data citation before the key finding. The paragraph should lead with the claim that Gemma 4 31B is the top dense open model, then support it with the table reference. This aligns with the standard "claim-evidence" structure expected in technical writing.

Finally, Section 4.2 contains a grammatical ambiguity where the participle "upholding" is loosely attached to the subject "We," creating a slight disconnect between the action (balancing speed) and the commitment (upholding the framework). A minor rephrasing to make the testing the subject of the "upholding" clause would resolve this.

Overall, the prose is competent but would benefit from tightening sentence structures and ensuring every paragraph opens with a clear, statable point.

Action items.

- [**writing**] The paper generally presents a clear narrative of the Gemma 4 model family, with a logical flow from architecture to training and evaluation. However, several sections suffer from structural issues that force the reader to re-parse sentences to identify the main point. In Section 2.1, the description of long-context efficiency is buried in a long, complex sentence that lists multiple architectural choices (sliding window ratios, p-RoPE, KV sharing) before stating the result (37.5% reduction). Th